

Subject : DS-505: Big Data Handling and Management for Machine Learning Applications
Course : Bachelors of Science in Data Science and Analytics Semester : Semester-5
Model Paper 3

Q.1 A	Answer the following in short (Any 4)	04 Marks
1	Which Python library provides high-performance data structures like Series and DataFrames for data manipulation? a) NumPy b) Pandas c) Matplotlib d) Scipy	
2	In PySpark, which function is used to filter rows based on a given boolean condition? a) df.select() b) df.filter() c) df.drop() d) df.distinct()	
3	Which Scikit-Learn function is used to scale features to have zero mean and unit variance ($\mu=0$, $\sigma=1$)? a) MinMaxScaler() b) StandardScaler() c) LabelEncoder() d) SimpleImputer()	
4	Which Python library developed by Hugging Face provides easy access to pretrained transformer models and pipelines? a) nltk b) spacy c) transformers d) gensim	
5	Which of the following is a Wide Transformation in PySpark that causes data shuffling across nodes? a) df.select() b) df.filter() c) df.groupBy() d) df.withColumn()	
Q.1 B	Answer the following in short (Any 4)	04 Marks
1	Mention the names of the 5 Vs of Big Data.	
2	Differentiate between an Action and a Transformation in PySpark.	
3	Differentiate between training data (X_{train}) and testing data (X_{test}).	
4	Name any two popular Large Language Models (LLMs) used today.	
5	Differentiate between Ordinal and Nominal categorical variables.	
Q.2 A	Write a short note on the following (any one)	05 Marks
1	Explain the concept of Memory Management while loading large datasets in Python. How does chunked reading with chunksize help in preventing memory crashes?	
2	How do you create and initialize a SparkSession in PySpark using SparkSession.builder? Write the Python code snippet and explain the importance of appName and master.	
B	Explain the concept of Lazy Evaluation in Apache Spark. Differentiate between Narrow Dependencies and Wide Dependencies (Shuffle) with appropriate examples.	04 Marks
Q.3	Write a short note on the following (any Two)	08 Marks
1	Explain the working of the Linear Regression model with a simple diagram. What is the difference between independent features and dependent target variable?	
2	What is a Large Language Model (LLM)? Explain any two popular real-world applications of LLMs (such as Chatbots and Text Summarization) with suitable examples.	
3	Explain the concept of Decision Tree Classifier with a simple diagram (Root Node, Decision Node, Leaf Node). How does a decision tree classify a new observation?	